Beta-Two Dyson–Ornstein–Uhlenbeck Diffusion: Exact Chamber Kernel and Noncollision

HCS-C378 theorem package

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Abstract

We fix trace-metric Hermitian Ornstein–Uhlenbeck diffusion with drift $-H/2$ and derive, without rescaling, unit eigenvalue noise, reciprocal-gap repulsion, and the ordered GUE reversible density. The Karlin–McGregor determinant for killed independent scalar Ornstein–Uhlenbeck particles has an exact Vandermonde mass. Its shifted Doob transform is therefore conservative, has the eigenvalue generator, is reversible, and never reaches a collision wall from an interior start. This second round closes the exact transition kernel and boundary nonattainment as a complete endpoint.

中文摘要

本文在迹度量归一化下研究贝塔二戴森奥恩斯坦乌伦贝克特征值扩散。
二阶矩阵扰动给出单位噪声、倒间隙排斥和线性回复，并确定有序高斯酉系综为可逆平衡分布。
为构造真正守恒的腔室转移核，先取相互独立的一维回复粒子在碰撞壁吸收的行列式核，
再证明其对范德蒙德函数的总质量恰好带有由次数决定的指数衰减。 补回这一能量位移后，杜布变换核质量为一，
其生成元逐项等于特征值生成元，并满足详细平衡。 由于该核在开腔室内给出全部有限时分布，
从内部出发的过程不会到达碰撞边界。 核的构造同时给出半群拼接和起始点极限， 但本文只主张从内部出发的非到达性，
不把它扩张成从碰撞壁起始的入口定律。 精确卡林麦格雷戈核、守恒变换与无碰撞定理共同构成第二轮的完整终点。

Keywords: Dyson diffusion; Karlin–McGregor kernel; Doob transform; Vandermonde function; noncollision; reversible diffusion
关键词：戴森扩散；卡林麦格雷戈核；杜布变换；范德蒙德函数；无碰撞；可逆扩散

1 One normalization from matrix to chamber

Let Herm_N carry the real trace inner product $\langle A, B\rangle = \mathrm{Tr}(AB)$. Standard Brownian motion $\mathcal{B}_t$ in this Euclidean space drives

$$\mathrm{d}H_t = \mathrm{d}\mathcal{B}_t - \frac{1}{2}H_t\,\mathrm{d}t. \tag{1}$$

The trace-orthonormal basis consists of E_{ii} and, for $i<j$,

$$\frac{E_{ij}+E_{ji}}{\sqrt{2}}, \qquad \frac{\mathrm{i}(E_{ij}-E_{ji})}{\sqrt{2}}. \tag{2}$$

Thus every diagonal matrix martingale in an instantaneous eigenbasis has quadratic variation $\mathrm{d}t$, while every off-diagonal complex entry has expected squared modulus $\mathrm{d}t$. This sentence fixes the factor that is often hidden in matrix-entry conventions.

Write

$$W_N = \{x\in\mathbb{R}^N : x_1<\cdots<x_N\}, \qquad h(x)=\prod_{i<j}(x_j-x_i), \qquad d=\frac{N(N-1)}{2}. \tag{3}$$

Theorem 1 (Radialization in the trace convention)**.** *If H_0 has simple spectrum, then, up to their first collision, its increasingly ordered eigenvalues solve*

$$\mathrm{d}X_i = \mathrm{d}B_i + \sum_{j\neq i}\frac{\mathrm{d}t}{X_i-X_j} - \frac{X_i}{2}\,\mathrm{d}t, \tag{4}$$

where the B_i are independent standard real Brownian motions. The collision time is infinite. The generator on W_N is

$$\mathcal{L} = \frac{1}{2}\sum_i \partial_i^2 + \sum_i\left(\sum_{j\neq i}\frac{1}{x_i-x_j} - \frac{x_i}{2}\right)\partial_i. \tag{5}$$

Proof. For a normalized eigenvector u_i, second-order Hermitian perturbation gives

$$\mathrm{d}X_i = \langle u_i,\, \mathrm{d}H\, u_i\rangle + \sum_{j\neq i} \frac{|\langle u_j,\, \mathrm{d}\mathcal{B}\, u_i\rangle|^2}{X_i - X_j}. \tag{6}$$

The covariance statement following (2) turns the first martingale term into $\mathrm{d}B_i$, the second-order term into the displayed reciprocal gaps, and the matrix drift into $-X_i\,\mathrm{d}t/2$. This proves the formula until the first collision. Section 2 proves that this time is infinite. □

Theorem 2 (Ordered GUE reversible law)**.** *The probability density*

$$\pi_N(x) = Z_N^{-1}\mathrm{e}^{-|x|^2/2}h(x)^2\,1_{W_N}(x), \qquad Z_N = (2\pi)^{N/2}\prod_{j=0}^{N-1} j!, \tag{7}$$

is reversible for (5)*.*

Proof. The drift is one half of the logarithmic gradient:

$$\frac{1}{2}\partial_i \log(\mathrm{e}^{-|x|^2/2}h^2) = -\frac{x_i}{2} + \sum_{j\neq i}\frac{1}{x_i - x_j}. \tag{8}$$

Hence $\mathcal{L}f = (2\pi_N)^{-1}\nabla\cdot(\pi_N\nabla f)$, where π_N denotes its density, proving symmetry. For the normalizer, use $h = \det[\mathrm{He}_{i-1}(x_j)]$ and scalar Hermite orthogonality in Andréief's identity. The integral over $\mathbb{R}^N$ is $N!(2\pi)^{N/2}\prod_{j=0}^{N-1} j!$. The integrand is symmetric, so the ordered chamber contributes $1/N!$ of that value. The invariant matrix density is proportional to $\mathrm{e}^{-\operatorname{Tr} H^2/2}$; Weyl integration gives the same ordered density. □

The $N = 1$ face is ordinary scalar Ornstein–Uhlenbeck diffusion. No large-N passage or entrance from a multiple eigenvalue is included.

2 Killed determinant and conservative Doob transform

The independent scalar generator is

$$\mathcal{L}_0 = \frac{1}{2}\sum_i(\partial_i^2 - x_i\partial_i). \tag{9}$$

For $t > 0$, put $r = \mathrm{e}^{-t/2}$ and $\sigma_t^2 = 1 - r^2$. Its Lebesgue density is

$$p_t(a,b) = \frac{1}{\sqrt{2\pi\sigma_t^2}}\exp[-\frac{(b-ra)^2}{2\sigma_t^2}]. \tag{10}$$

Lemma 1 (Vandermonde energy)**.** *The polynomial h is harmonic and homogeneous of degree d. Consequently,*

$$\mathcal{L}_0 h = -\frac{d}{2}h. \tag{11}$$

Proof. The permutation-invariant Laplacian sends the alternating polynomial h to an alternating polynomial of degree $d-2$. Every nonzero alternating polynomial is divisible by h, which already has degree d; hence $\Delta h = 0$. Euler's homogeneous identity gives $x\cdot\nabla h = dh$. Substitution in (9) proves (11). □

Theorem 3 (Exact chamber kernel and boundary)**.** *The independent process killed at its first collision has density*

$$q_t(x,y) = \det[p_t(x_i,y_j)]_{i,j=1}^N. \tag{12}$$

Its Vandermonde mass is

$$\int_{W_N} q_t(x,y)h(y)\,\mathrm{d}y = r^d h(x). \tag{13}$$

Therefore

$$k_t(x,y) = r^{-d}\frac{h(y)}{h(x)}q_t(x,y) \tag{14}$$

is a conservative Markov density with generator (5)*. Starting from any $x\in W_N$, its continuous diffusion never reaches a collision wall in finite time.*

Proof. The reflection principle pairs every wall-hitting independent path with the path reflected after its first hit. The signed permutation sum that remains is the Karlin–McGregor determinant (12).

Both $q_t(x,\cdot)$ and h are alternating. Their product is symmetric, so the chamber integral in (13) is $1/N!$ of the full-space integral. Write $h(y) = \det[y_j^{i-1}]$. Andréief's identity gives

$$\int_{\mathbb{R}^N} q_t(x,y)h(y)\,\mathrm{d}y = N!\det\Big[\int_{\mathbb{R}} p_t(x_i,y)y^{j-1}\,\mathrm{d}y\Big]_{i,j=1}^N. \tag{15}$$

The $(j-1)$st moment of $N(rx_i, 1-r^2)$ is a degree-$(j-1)$ polynomial in x_i with leading coefficient r^{j-1}. Lower-degree column operations reduce the determinant in (15) to $r^{0+\cdots+(N-1)}h(x) = r^d h(x)$. This proves (13).

Equation (13) makes (14) conservative. On smooth compactly supported functions,

$$h^{-1}\mathcal{L}_0(hf) + \frac{d}{2}f = \mathcal{L}_0 f + \nabla \log h \cdot \nabla f = \mathcal{L}f, \tag{16}$$

because $\partial_i \log h = \sum_{j\neq i}(x_i - x_j)^{-1}$. The minimal transformed process has mass one at every finite time. A boundary hit by a continuous chamber path would be its killing time, so conservativity excludes such a hit. Local uniqueness before collision and the standard electrostatic-repulsion theorem identify the transform with the unique global strong solution of (4). □

Scalar detailed balance in (10) passes through the determinant; the two Vandermonde factors in (14) then prove $\pi_N(x)k_t(x,y) = \pi_N(y)k_t(y,x)$ directly. This independently checks the generator calculation.

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round one determinant Doob kernel and no collision